

DP 301: Inter Socio Technical Practicum (ISTP) / Free Elective

Course Code	DP 301
Course Name	Inter Socio Technical Practicum (ISTP) / Free Elective
L-T-P-C	4 credits
Offered in	B.Tech (all branches) - Institute Core; B.Tech General Engineering; B.Tech General Engineering (Open Specialisation); B.Tech General Engineering with Data Analytics for Management & Finance (DAMF); B.Tech Microelectronics and VLSI Engineering
Approval	39th Senate (Apr 2023), in the B.Tech Microelectronics and VLSI Engineering curriculum; 44th Senate (Sep 2024)

This course follows the syllabus of DP 301P, reproduced below.

Course Code	DP 301P
Course Name	Interactive Socio-Technical Practicum
L-T-P-C	0-0-8-4
Prerequisites	Consent of the faculty member
Intended for	B.Tech
Distribution	Core Equivalent course: IC 301 (DP 301 and IC 301 are same)
Approval	2nd Adhoc Senate (May 2012)
Revised	13th Senate (Jun 2017) - renamed from "Inter-disciplinary Socio-Technical Practicum"

Course Contents

In the course the teams have two choices. Either they work iteratively on the earlier developed prototype to develop into final improved product or identify a new product which the society needs after doing market research.

In a changing market, staying competitive often requires the development of new products. As user requirements and needs change, products must also change. Market research is an essential tool to help boost the chances for success. The new product development process requires information from the market and users as to what is needed to support critical decisions about the product.

Key Learning Topics

- **Primary Research:** Information collection through various channels such as interviews, questionnaires, surveys, and conversations with industry experts, prospective customers, and competitors
- **Secondary Research:** Internal source (Brainstorming, Stock analysis, Retail data, loyalty cards etc.); External source (Government Statistics like ONS, Trade publications, Commercial Data, Household Expenditure Survey, Magazine surveys, Research documents like publications, journals, etc.)
- **Sampling Methods:** Random Samples, Stratified or Segment Random Sampling, Quota Sampling, Cluster Sampling, Multi-Stage Sampling, Snowball Sampling.
- **Market Trends:** Size of market, market trends, forecasting, planning, identify market strategies, identify user need, identify competition, identifying opportunities/gaps in market
- **Module V:** Engineering economics of the product

References

1. Von Hippel, Eric., *The sources of Innovation*, Oxford University Press, 1988.
2. Gordon, William., *The development of Creative Capacity*, Collier Books, 1961.
3. Thomke, Stefan, and Eric Von Hippel, *Customers as Innovators: A new way to create value*, Harvard Business Review (April 2005), 74-81, Reprint no. R 0304 F.
4. Boyd, Harper W. Jr., Westfall, Ralph and Stasch, Stanley, *Marketing Research: Text and Cases*, Richard D. Irwin Inc.
5. Green, P. E. and Tull, D. S., *Research for Marketing Decisions*, 5th edition, Prentice-Hall of India.
6. Luck D. J., Wales, H.G., Taylor, D. A. and Rubin R. S., *Marketing Research*, 7th Edition, Prentice-Hall of India.
7. Tull, D. S. and Hawkins D. I., *Marketing Research: Measurement and Method*, 6th Edition, Prentice-Hall of India.